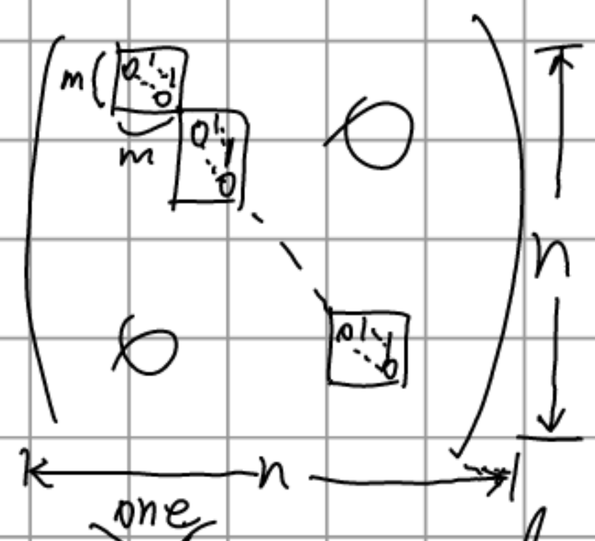
(JNF)

Case 1: $\chi_T(x) = x^n$, $m_T(x) = x^m$

$\Rightarrow \exists$ basis B of V,

s.t. $(T)_{B,B} =$



There is always $m \times m$ Jordan block

($\exists$)

The sizes of other bloks:

found by $\dim(\ker(T^i))$, $1 \le i \le m$

(见本周Tut)

The number of blocks (all) $= \dim(\ker(T))$

Case 2: $m_T(x) = (x-\mu)^m$

Consider $S: V \to V$,

where $S := T - \mu Id$

Then $m_S(x) = x^m$

$\Rightarrow$ Case I gives:

$$(S)_{B,B} = \begin{pmatrix} m\left(\begin{smallmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{smallmatrix}\right) & & 0 \\ & \ddots & \\ 0 & & \begin{smallmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{smallmatrix} \end{pmatrix} = (T)_{B,B} - \mu (I)_{B,B}$$

$$\Rightarrow (T)_{B,B} = \begin{pmatrix} m\left(\begin{smallmatrix} \mu & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \mu \end{smallmatrix}\right) & & 0 \\ & \ddots & \\ 0 & & \begin{smallmatrix} \mu & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \mu \end{smallmatrix} \end{pmatrix}$$

Case 3: $m_T(x) = (x-\mu_1)^{e_1} \cdots (x-\mu_k)^{e_k}$

Apply PDT:

where $\begin{cases} V_i = \ker (T-\mu_i Id)^{e_i} \\ V_i \text{ are } T\text{-invariant} \\ \text{For } T|_{V_i}: V_i \to V_i,\ m_{T|_{V_i}}(x) = (x-\mu_i)^{e_i} \end{cases}$

$V = V_1 \oplus \cdots \oplus V_i \oplus \cdots \oplus V_k$

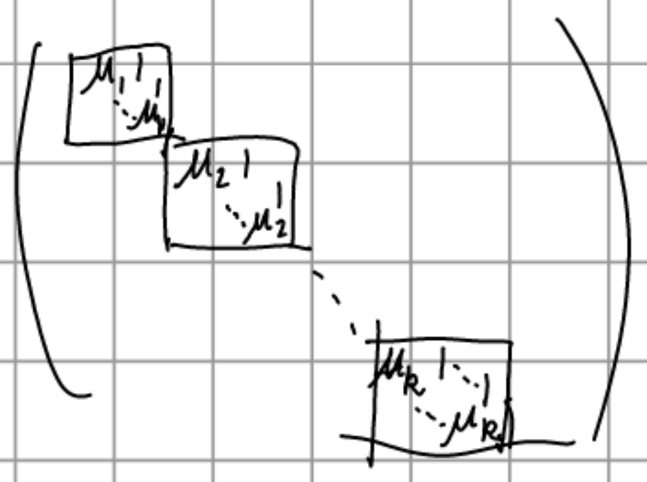
$\downarrow$ basis

$B := B_1 \cup \cdots \cup B_i \cup \cdots \cup B_k$

By $\curvearrowright$, $(T)_{B,B} = \begin{array}{c} \\ V_1 \\ V_2 \\ \vdots \\ V_k \end{array} \begin{pmatrix} \overbrace{\quad}^{V_1} & \overbrace{\quad}^{V_2} & & \overbrace{\quad}^{V_k} \\ M_1 & & & \\ & M_2 & & \\ & & \ddots & \\ & & & M_k \end{pmatrix}$,

where $M_i = (T|_{V_i})_{B_i,B_i}$ with $m_{T|_{V_i}}(x) = (x-\mu_i)^{e_i}$

By Case 2,

choose B_i s.t. $M_i = \sim =$ 

Inner Product Space

(Goal : generalize dot product in $\mathbb{R}^n$)

Bilinear form for $\mathbb{R}$-vector space in V

Def.: V: a vector space over $\mathbb{R}$

A bilinear form on V: a mapping $F: V \times V \to \mathbb{R}$

1. $F(\underline{u}+\underline{v}, \underline{w}) = F(\underline{u}, \underline{w}) + F(\underline{v}, \underline{w})$
2. $F(\underline{u}, \underline{v}+\underline{w}) = F(\underline{u}, \underline{v}) + F(\underline{u}, \underline{w})$
3. $F(\lambda \underline{u}, \underline{v}) = F(\underline{u}, \lambda \underline{v}) = \lambda F(\underline{u}, \underline{v})$

E.g.: Write 3. as $\langle \lambda \underline{u}, \underline{v} \rangle = \langle \underline{u}, \lambda \underline{v} \rangle = \lambda \langle \underline{u}, \underline{v} \rangle$ $(\lambda \in \mathbb{R})$

E.g.: (a) $V = \mathbb{R}^n$, $\langle \underline{u}, \underline{v} \rangle := \underline{u}^t \underline{v} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}^t \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$

(usual dot product)

$= (u_1 \cdots u_n) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = u_1 v_1 + \cdots + u_n v_n$

↑ $(p=n, q=0)$

(b) $V = \mathbb{R}^n$, $n = p+q$,

$\langle \underline{u}, \underline{v} \rangle_{p,q} := (u_1 v_1 + \cdots + u_p v_p) - (u_{p+1} v_{p+1} \cdots u_n v_n)$

(Ex.: $\langle \ , \ \rangle_{p,q}$ satisfies 1. ~ 3. in ☆)

(c) $V = C^\infty([0, 1])$

For $f, g \in C^\infty(\mathbb{R})$, $\langle f, g \rangle := \int_0^1 f(x) g(x) dx$

$\Rightarrow \langle f, f \rangle = \int_0^1 f^2(x) dx \geq 0 \Rightarrow$ positive definite

② Def.: $\langle \cdot, \cdot \rangle$

is symmetric

$\Leftarrow_{(if)} \langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle$

is positive definite ... ①

$\Leftarrow \begin{cases} \forall \underline{v}, \langle \underline{v}, \underline{v} \rangle \geq 0 \\ \langle \underline{v}, \underline{v} \rangle = 0 \Leftrightarrow \underline{v} = \underline{0} \end{cases}$

is nondegenerate

$\Leftarrow \left(\langle \underline{u}, \underline{v} \rangle = 0, \forall \underline{v} \Rightarrow \underline{u} = \underline{0} \right)$

Remark:

positive definite $\Rightarrow$ nondegenerate

Proof:

Suppose $\underline{u}$ satisfies $\forall \underline{v} \in V, \langle \underline{u}, \underline{v} \rangle = 0$

Then take $\underline{v} = \underline{u} \Rightarrow \langle \underline{u}, \underline{u} \rangle = 0 \Leftrightarrow \underline{u} = 0$

(Bilinear form for $\mathbb{C}$-v. space)

We can define $\langle , \rangle$ bilinear on $\mathbb{C}$-v. space V

(12) Such form is never positive definite

證: Suppose $\langle \underline{v}, \underline{v} \rangle > 0$

Then $\langle i\underline{v}, i\underline{v} \rangle = (-1)\langle \underline{v}, \underline{v} \rangle < 0$

Sesqui-linear Form

V: a vector space over $\mathbb{C}$

A bilinear form on V: a mapping $F: V \times V$

1. $F(\underline{u}+\underline{v}, \underline{w}) = F(\underline{u}, \underline{w}) + F(\underline{v}, \underline{w})$
2. $F(\underline{u}, \underline{v}+\underline{w}) = F(\underline{u}, \underline{v}) + F(\underline{u}, \underline{w})$
3. $F(\overline{\lambda}\underline{u}, \underline{v}) = F(\underline{u}, \lambda\underline{v}) = \lambda F(\underline{u}, \underline{v})$

$\langle \lambda\underline{u}, \underline{v} \rangle = \overline{\lambda}\langle \underline{u}, \underline{v} \rangle$

(and)

$\langle \underline{u}, \lambda\underline{v} \rangle = \lambda\langle \underline{u}, \underline{v} \rangle$

↓

Suppose $\langle \underline{v}, \underline{v} \rangle > 0$

Then $\langle i\underline{v}, i\underline{v} \rangle = (-i \cdot i)\langle \underline{v}, \underline{v} \rangle = 1\langle \underline{v}, \underline{v} \rangle > 0$

↳ is Conjugate-symmetric

$\Leftarrow\!= \langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle$

E.g.: $V = \mathbb{C}^n$, $\langle \underline{u}, \underline{v} \rangle := \overline{u_1} v_1 + \cdots + \overline{u_n} v_n$

Inner product space

Def: ↓ For $\mathbb{R}$ – v. space:
bilinear, symmetric, positive definite

↓ For $\mathbb{C}$ – v. space:
sesqui-linear, conjugate symmetric,
positive definite

Let $(V, \langle , \rangle)$ be an inner product space

(1) $\|\underline{v}\| := \sqrt{\langle \underline{v}, \underline{v} \rangle}$ is the <u>norm of $\underline{v}$</u>

(2) $\|\underline{v}\| = 1 \iff \underline{v}$ is <u>unit vector</u>

(3) $\underline{u}, \underline{v}$ are orthogonal ($\underline{u} \perp \underline{v}$)
$\iff \langle \underline{u}, \underline{v} \rangle = 0$

Prop. (Gram-Schmidt process):

$\{\underline{v}_1, \cdots, \underline{v}_k\}$ is linear indep.

$\Rightarrow \exists \{\underline{w}_1, \cdots, \underline{w}_k\}$,

s.t. $\forall i, \|\underline{w}_i\| = 1$,

$\forall i \neq j, \langle \underline{w}_i, \underline{w}_j \rangle = 0$,

$\mathrm{Span}(\{\underline{v}_1, \cdots, \underline{v}_k\}) = \mathrm{Span}(\{\underline{w}_1, \cdots, \underline{w}_k\})$

given by:

$$\underline{w}_1 = \frac{\underline{v}_1}{\|\underline{v}_1\|}$$

and $\forall i$, $\underline{w}_{i+1} = \dfrac{\underline{w}'_{i+1}}{\|\underline{w}'_{i+1}\|}$,

where $\underline{w}'_{i+1} := \underline{v}_{i+1} - \left(\langle \underline{v}_{i+1}, \underline{w}_1 \rangle \underline{w}_1 + \cdots + \langle \underline{v}_{i+1}, \underline{w}_i \rangle \underline{w}_i \right)$

Orthogonal Complement

Def.: Let $U \subseteq V$ be a subset

Then (the $\perp$ of U:) $U^\perp = \{\underline{v} \in V \mid \langle \underline{v}, \underline{u} \rangle = 0, \forall \underline{u} \in U\}$

Prop.: (a) $U^\perp \leq V$ is a vector subspace

(b) $U \subseteq (U^\perp)^\perp$

(c) $\left.\begin{array}{l}\dim(V)<\infty \\ W \leq V\end{array}\right\} \Rightarrow \left\{\begin{array}{l}\dim(W)+\dim(W^\perp)=\dim(V) \\ W \oplus W^\perp = V\end{array}\right.$

Proof:

(a) Let $\underline{w}_1, \underline{w}_2 \in U^\perp$

$\Rightarrow \forall \underline{u} \in U$, $\langle a\underline{w}_1 + b\underline{w}_2, \underline{u} \rangle$

$= \langle a\underline{w}_1, \underline{u} \rangle + \langle b\underline{w}_2, \underline{u} \rangle$

$= \bar{a}\langle \underline{w}_1, \underline{u} \rangle + \bar{b}\langle \underline{w}_2, \underline{u} \rangle$

$= 0 + 0 = 0$

(b) Let $\underline{w} \in U$

$\forall \underline{x} \in U^\perp$, $\langle \underline{w}, \underline{x} \rangle = 0$

$\Rightarrow \underline{w} \in (U^\perp)^\perp$

(c) Let $\{\underline{w}_1, \cdots, \underline{w}_k\}$ be a basis of W

basis extension $\rightarrow$ $\{\underline{w}_1, \cdots, \underline{w}_k, \underline{v}_{k+1}, \sim, \underline{v}_n\}$ be a basis of V

Gram-Schmidt $\rightarrow$ $\{\underbrace{\underline{u}_1, \cdots, \underline{u}_k}_{\text{orthogonal basis of } W}, \underline{u}_{k+1}, \cdots, \underline{u}_n\}$ be an orthogonal basis of V

(HW: $W^\perp = \text{Span}\{\underline{u}_{k+1}, \cdots, \underline{u}_n\}$)